Summary in Graph

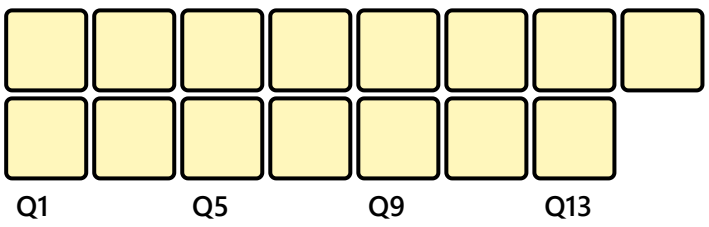
Exam Summary (GO Classes GATE CS/DA | DBMS | Normalization, Relational Model | Weekly Quiz 3)

Qs. Attempted:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)
Correct Marks:	0 (0 + 0)
Penalty Marks:	0 (0 + 0)
Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (68) out of 683 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Databases

Consider the following relation state for $R(A, B, C)$

```
R
+----+----+----+
| A  | B  | C  |
+----+----+----+
| 10 | b1 | c1 |    (row 1)
| 10 | b1 | c2 |
| 10 | b1 | c1 |
| 12 | b3 | c4 |
| 10 | b1 | c1 |
| 12 | b3 | c4 |
+----+----+----+
```

Which of the following dependencies may hold in the above relation?

A. $AB \rightarrow C$
B. $C \rightarrow AB$
C. $B \rightarrow C$
D. $AC \rightarrow B$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Which of the following statements best describes the main reason for representing a relational database in BCNF?

A. To achieve physical data independence.
B. To remove data anomalies (insertion, update, deletion anomalies).
C. To save space on disk.
D. To achieve logical data independence.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Which of the following statements is/are true?

1. Each superkey is a superset of some candidate key.
2. Each primary key is also a candidate key, but there may be candidate keys that are not primary keys.

A. only 1 is true
B. only 2 is true
C. both 1 and 2 are true
D. neither 1 nor 2 are true

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider the relational scheme $\mathrm{R(A, B, C, D, E, F, G, H)}$ and the following functional dependencies over

$$\mathrm{R : A \rightarrow BCD; AD \rightarrow E; EFG \rightarrow H; F \rightarrow GH}$$

Based on these functional dependencies, there is one key for R. One of these four functional dependencies can be removed without altering the key. Which one?

A. $\mathrm{A \rightarrow BCD}$
B. $\mathrm{AD \rightarrow E}$

C. $\mathrm{EFG} \rightarrow \mathrm{H}$
D. $\mathrm{F} \rightarrow \mathrm{GH}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider the relational schema $\mathrm{R}(\mathrm{A}\ \mathrm{B}\ \mathrm{C}\ \mathrm{D})$ with following functional dependency set $F = \{A \rightarrow BC, C \rightarrow D\}$; The relation R is in

A. 2 NF
B. BCNF
C. 3 NF
D. 1 NF

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the relation $\mathbf{R}(\underline{\mathbf{A}, \mathbf{B}}, \mathbf{C}, \mathbf{D}, \mathbf{E}, \mathbf{F})$ with $\{\mathrm{A}, \mathrm{B}\}$ as a composite primary key and the following dependencies:

- $\{\mathrm{A}, \mathrm{B}\} \rightarrow \mathrm{C}, \mathrm{D}, \mathrm{E}, \mathrm{F}$
- $\mathrm{E} \rightarrow \mathrm{D}$
- $\mathrm{C} \rightarrow \mathrm{F}$

Which of the following options are true? (Select all correct choices)

A. R is in $3\mathrm{NF}$ (i.e. 3rd Normal Form)
B. R is in $2\mathrm{NF}$ (i.e. 2nd Normal Form)
C. R is in $1\mathrm{NF}$ (i.e. 1st Normal Form)
D. R is in $2\mathrm{NF}$ but not in $3\mathrm{NF}$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the following relational schema and set of functional dependencies. $R(A, B, C, D, E)$ with functional dependencies $AB \rightarrow E$ and $D \rightarrow C$. Let X be the number of superkeys for this relation. Let Y be the number of minimal superkeys for this relation. What is $\mathrm{X} + \mathrm{Y}$?

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider a Relation $\mathrm{R}(\mathrm{ABCD})$. For which of the following set of functional dependencies (FDs), relation R is in Boyce-Codd Normal Form (BCNF)?

A. $\mathrm{BD} \rightarrow \mathrm{AC}; \mathrm{AB} \rightarrow \mathrm{D}; \mathrm{AC} \rightarrow \mathrm{B}$
B. $\mathrm{A} \rightarrow \mathrm{C}; \mathrm{B} \rightarrow \mathrm{AC}; \mathrm{C} \rightarrow \mathrm{D}.$
C. $\mathrm{A} \rightarrow \mathrm{BCD}; \mathrm{B} \rightarrow \mathrm{ACD}$

D. $\mathrm{BC} \rightarrow \mathrm{AD}; \mathrm{D} \rightarrow \mathrm{B}$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Suppose R and S are two relations. Let C be the referential integrity constraint that says: whenever R has a tuple with some values $v_1, v_2, \ldots, v_n$ in particular attributes $A_1, A_2, \ldots, A_n$, there must be a tuple of S that has the same values $v_1, v_2, \ldots, v_n$ in particular attributes $B_1, B_2, \ldots, B_n$.

Which of the following operations on R, S can cause violation of the constraint C?

A. Insert into R
B. Insert into S
C. Delete from R
D. Delete from S

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Which one of the following statements about normal forms is FALSE?

A. BCNF is stricter than $3\mathrm{NF}$
B. Lossless, dependency-preserving decomposition into $3\mathrm{NF}$ is always possible
C. Lossless, dependency-preserving decomposition into BCNF is always possible
D. Any relation with two attributes is in BCNF

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following functional dependencies in a database:

- $A \rightarrow B$
- $B \rightarrow C$
- $D \rightarrow E$
- $E \rightarrow D$
- $F \rightarrow G$
- $F \rightarrow H$
- $(E, F) \rightarrow I$

The relation (E, D, A, B) is

A. 2 NF
B. 3 NF
C. BCNF
D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the functional dependencies (FDs) of a relation R and below-mentioned properties:

- $\mathrm{P1}$: A non-prime attribute determines a different non-prime attribute.
- $\mathrm{P2}$: A proper subset of a candidate key determines a non-prime attribute.

Choose the correct option:

A. If some FD of R has property $\mathrm{P1}, \mathrm{R}$ can't be in $2\mathrm{NF}$
B. If some FD of R has property $\mathrm{P2}, \mathrm{R}$ can't be in $2\mathrm{NF}$
C. $\mathrm{P1}$ and $\mathrm{P2}$ won't affect R being in $2\mathrm{NF}$
D. If some FD of R has property $\mathrm{P2}, \mathrm{R}$ can still be in $2\mathrm{NF}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Following table has two attributes Employee_id and Manager_id, where Employee_id is a primary key and manager_id is a foreign key referencing Employee_id with on-delete cascade:

Employee_id	Manager_id
20	40
25	40
30	35
35	20
40	45
45	25

On deleting the table $(20, 40)$, the set of other tuples that must be deleted to maintain the referential integrity of table is

A. $(30, 35)$ only
B. $(30, 35)$ and $(35, 20)$ only
C. $(35, 20)$ only
D. $(40, 45)$ and $(25, 40)$ only

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following relational schema and set of functional dependencies.

$\mathrm{R}(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E})$ with functional dependencies $\mathrm{ABC} \rightarrow \mathrm{D}; \mathrm{CD} \rightarrow \mathrm{A}; \mathrm{AD} \rightarrow \mathrm{E}$.

Which of the following is true?

A. R is in $1\mathrm{NF}$ but not in $2\mathrm{NF}$.
B. R is in $2\mathrm{NF}$ but not in $3\mathrm{NF}$.
C. R is in $3\mathrm{NF}$ but not in BCNF.
D. R is in BCNF.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

An attribute A is called prime if A is in any of the candidate keys. Assume in the following statements, A is an attribute & $\mathcal{X}$ is a set of attributes.

Consider the following statements:

A. Domain of each attribute is an elementary type; that is, not a set or a record structure.

B. Whenever $\mathcal{X} \mapsto A$ is a functional dependency that holds in relation $\mathbf{R}$ and $A \notin \mathcal{X}$, then either
- A is prime, or
- $\mathcal{X}$ is a key or a super-key for $\mathbf{R}$.

C. Whenever $\mathcal{X} \mapsto A$ is a functional dependency that holds in relation $\mathbf{R}$ and $A \notin \mathcal{X}$, then either
- A is prime, or
- $\mathcal{X}$ is not a proper subset of any key for $\mathbf{R}$.

D. Whenever $\mathcal{X} \mapsto A$ is a functional dependency that holds in relation $\mathbf{R}$ and $A \notin \mathcal{X}$, then
- $\mathcal{X}$ is a key or a super-key for $\mathbf{R}$.

Which of the following is a correct match??

A. $\mathrm{A} - 1\mathrm{NF}, \mathrm{B} - 2\mathrm{NF}, \mathrm{C} - 3\mathrm{NF}, \mathrm{D} - \mathrm{BCNF}$
B. $\mathrm{A} - 1\mathrm{NF}, \mathrm{B} - \mathrm{BCNF}, \mathrm{C} - 2\mathrm{NF}, \mathrm{D} - 3\mathrm{NF}$
C. $\mathrm{A} - 1\mathrm{NF}, \mathrm{B} - 3\mathrm{NF}, \mathrm{C} - 2\mathrm{NF}, \mathrm{D} - \mathrm{BCNF}$
D. $\mathrm{A} - \mathrm{BCNF}, \mathrm{B} - 3\mathrm{NF}, \mathrm{C} - 2\mathrm{NF}, \mathrm{D} - 1\mathrm{NF}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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